post-CMOS Design Rule

Version Record

No.	Version	Description	Project	Owner	Date
1	0.1	Initiate	Silicon Interposer Package	XXX	2025/XX/X
2					
3					
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1 Introduction

This document provides the layer information for the design and design rule about layout of integration circuits for silicon interposer Package process technology.

2 General Layout Information

This chapter provides the following general layout information:

2.1 Mask Layer

2.2 Layer mapping

2.3 Cross section

2.1 Mask Information

2.1.1 ~~Mask Layer~~

2.1.2 Layer Usage

Layer Name	Layer Type	Gds Number	Datatype (texttype)	Description
VIA0	drawing	101	0	
DVIA0	drawing	102	0	
TSV	drawing	103	0	tsv for connect M1 and BM1
TSV_AUX	drawing	104	1	Auxiliary layer for tsv device
TSVlabel	drawing	105	2	Label layer for TSV
M1	drawing	106	0	Metal1 conductor layer
M1dummy	drawing	107	1	Dummy layer for metal1
M1label	text	108	2	Label layer for metal1
VIA1	drawing	109	0	Via1 for metal1 and metal2
M2	drawing	110	0	Metal2 conductor layer
M2dummy	drawing	111	1	Dummy layer for metal2
M2label	text	112	2	Label layer for metal2
VIA2	drawing	113	0	Via2 for metal2 and metal3
M3	drawing	114	0	Metal3 conductor layer
M3dummy	drawing	115	1	Dummy layer for metal3
M3label	text	116	2	Label layer for metal3
DM1	drawing	117	0	Damascus process metal1
DM1dummy	drawing	118	1	Damascus process metal1 dummy
DM1label	text	119	2	Damascus process metal1 label
DVIA1	drawing	120	0	Damascus process Via1
DM2	drawing	121	0	Damascus process metal2
DM2dummy	drawing	122	1	Damascus process metal2 dummy
DM2label	text	123	2	Damascus process metal2 label
DVIA2	drawing	124	0	Damascus process Via2
DM3	drawing	125	0	Damascus process metal3
DM3dummy	drawing	126	1	Damascus process metal3 dummy
DM3label	text	127	2	Damascus process metal3 label
RV	drawing	128	0	RV layer for UBM
UBM	drawing	129	0	UBM layer
UBMlabel	text	130	2	UBM layer label
RVwafer	drawing	131	0	RV layer for wafer2wafer process
UBMwafer	drawing	132	0	UBM layer for wafer2wafer process

UBMwaferlabel	text	133	2	UBM label layer for wafer2wafer process
BM1	drawing	134	0	Metal1 layer for backside
BM1dummy	drawing	135	1	Metal1 dummy layer for backside
BM1label	text	136	2	Metal1 label layer for backside
BVIA1	drawing	137	0	Via1 layer for backside
BM2	drawing	138	0	Metal2 layer for backside
BM2dummy	drawing	139	1	Metal2 dummy layer for backside
BM2label	text	140	2	Metal2 label layer for backside
BVIA2	drawing	141	0	Via2 layer for backside
BM3	drawing	142	0	Metal3 layer for backside
BM3dummy	drawing	143	1	Metal3 dummy layer for backside
BM3label	text	144	2	Metal3 label layer for backside
DBM1	drawing	145	0	Metal1 layer for backside Damascus
DBM1dummy	drawing	146	1	Metal1 dummy layer for backside Damascus
DBM1label	text	147	2	Metal1 label layer for backside Damascus
DBVIA1	drawing	148	0	Via1 layer for backside Damascus
DBM2	drawing	149	0	Metal2 layer for backside Damascus
DBM2dummy	drawing	150	1	Metal2 dummy layer for backside Damascus
DBM2label	text	151	2	Metal2 label layer for backside Damascus
DBVIA2	drawing	152	0	Via2 layer for backside Damascus
DBM3	drawing	153	0	Metal3 layer for backside Damascus
DBM3dummy	drawing	154	1	Metal3 dummy layer for backside Damascus
DBM3label	text	155	2	Metal3 label layer for backside Damascus
BRV	drawing	156	0	RV layer for backside
BUBM	drawing	157	0	UBM layer for backside
BUBMlabel	text	158	2	UBM label layer for backside
BRVwafer	drawing	159	0	RV layer for backside wafer2wafer process
BUBMwafer	drawing	160	0	UBM layer for backside wafer2wafer process
BUBMwaferlabel	text	161	2	UBM label layer for backside wafer2wafer process
Die	drawing	162	0	Die frame layer for package layout design
DiePad	drawing	163	0	Pad layer of Die for connecting UBM layer
DiePadlabel	text	164	2	Recognize die pad label for LVS check
ALPAD	drawing	165	0	Auxiliary layer for connectivity between TSV and VIA0
ALPADlabel	text	166	2	Recognize ALPAD label for LVS check

2.1.3 Metal Option

1, Organic process and Damascus process

Metal option of silicon front:

Metal option	1M	2M	3M
ALPAD	V	V	V
VIA0/DVIA0	V	V	V
M1/DM1	V	V	V
VIA1/DVIA1		V	V

M2/DM2		V	V
VIA2/DVIA2			V
M3/DM3			V

Metal option of silicon backside:

Metal option	1M	2M	3M
BM1/DBM1	V	V	V
BVIA1/DBVIA1		V	V
BM2/DBM2		V	V
BVIA2/DBVIA2			V
BM3/DBM3			V

The interposer front metal option and backside can freely combine, example as front 3M & backside 1M.

The frontend process include Organic process(Mn/ VIAn) and Damascus process(DMn/ DVIAn). The backside process include Organic process(BMn/ BVIAn) and Damascus process(DBMn/ DBVIAn). Only one process of Organic or Damascus allowed in one chip design.

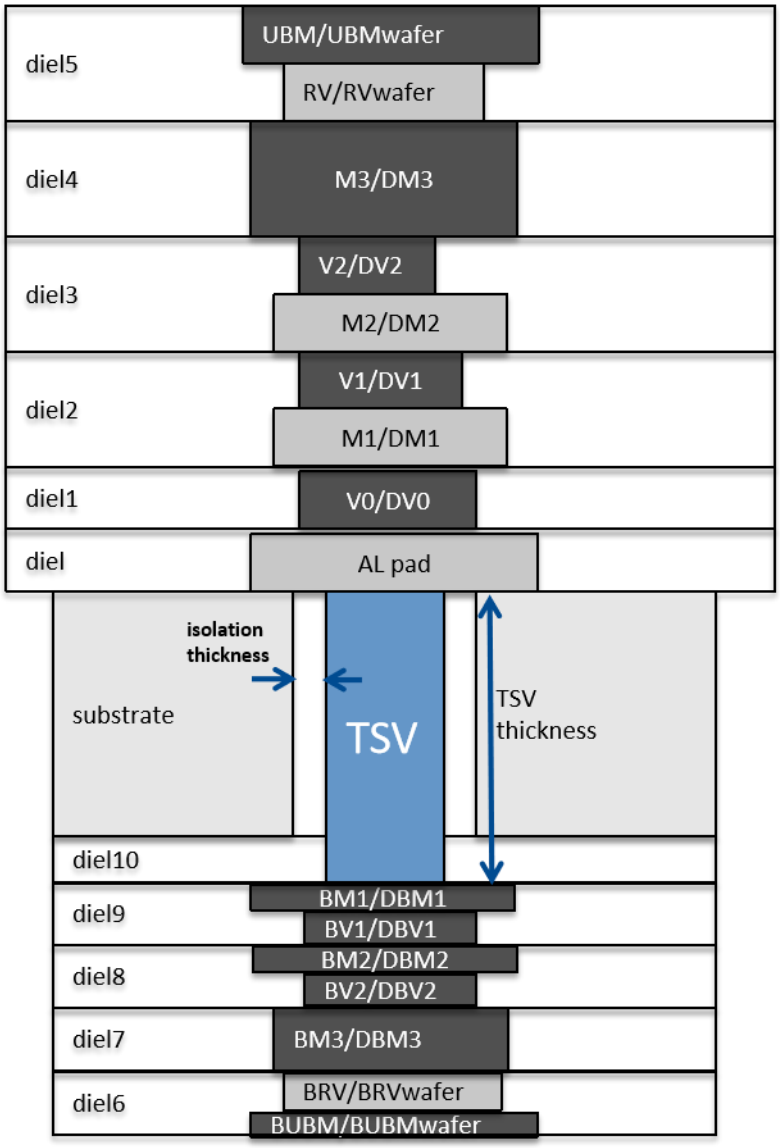
2, WLP process

Metal option	1M	2M	3M
ALPAD	V	V	V
VIA0	V	V	V
M1	V	V	V
VIA1		V	V
M2		V	V
VIA2			V
M3			V

WLP process only have Mn/VIAn metal option, it mean only have silicon front RDL.

2.2 Cross-section

1, Organic process and Damascus process



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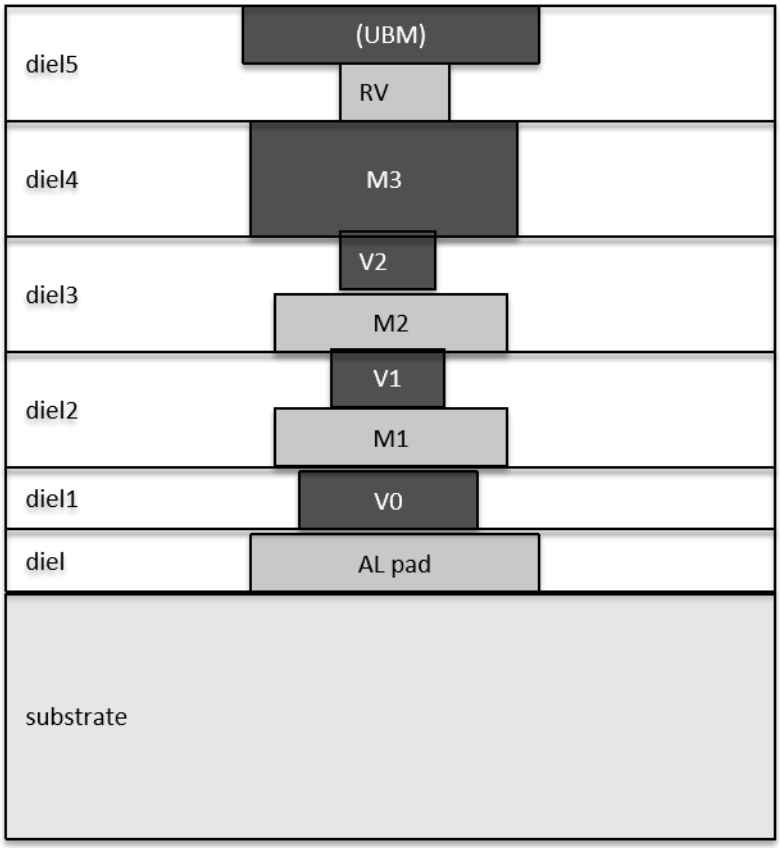
The front ubump include UBM and UBMwafer, back ubump include BUBM and BUBMwafer. For Organic process, only UBM and BUBM can combination. For Damascus process, the front can be UBM and UBMwafer, the back can be BUBM and BUBMwafer, the interposer front and backside can freely combine, example as front UBM & backside BUBMwafer

2, WLP process



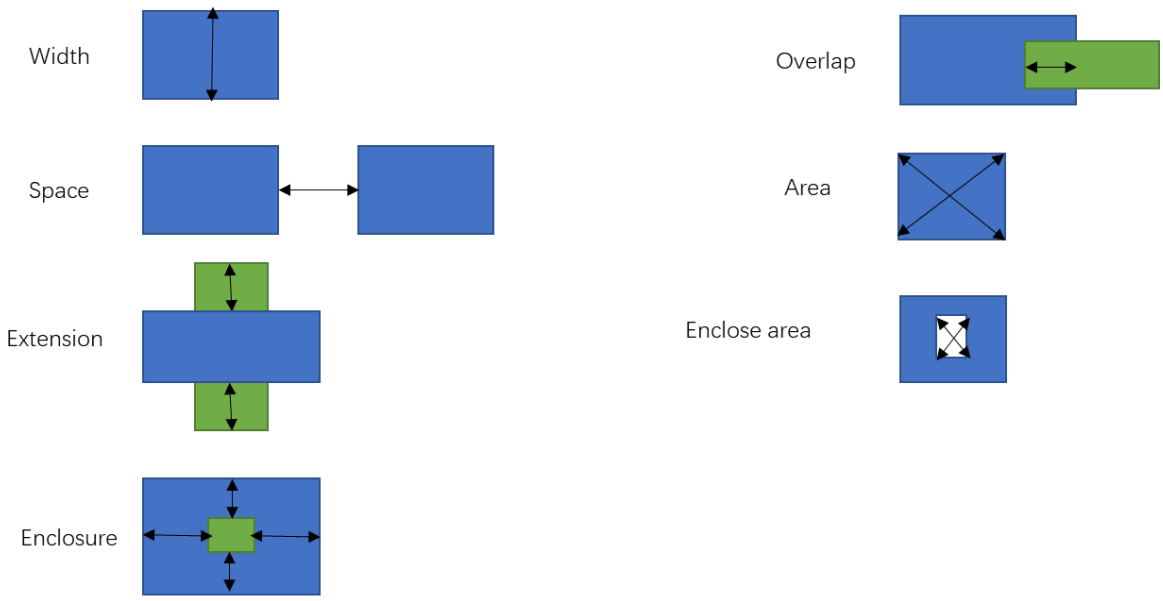
WLP process only have Mn/VIAn metal option, it mean only have silicon front RDL

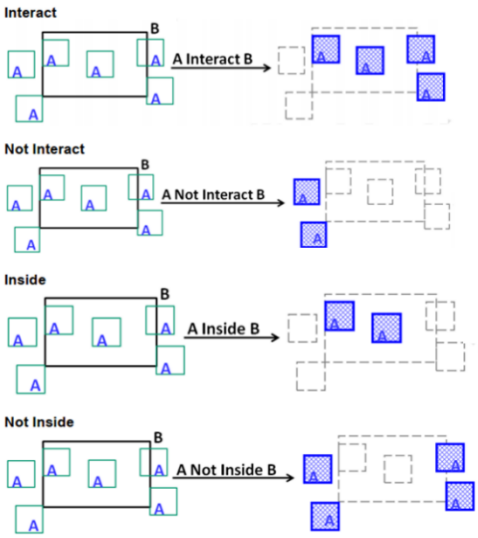
3 Layout Rules

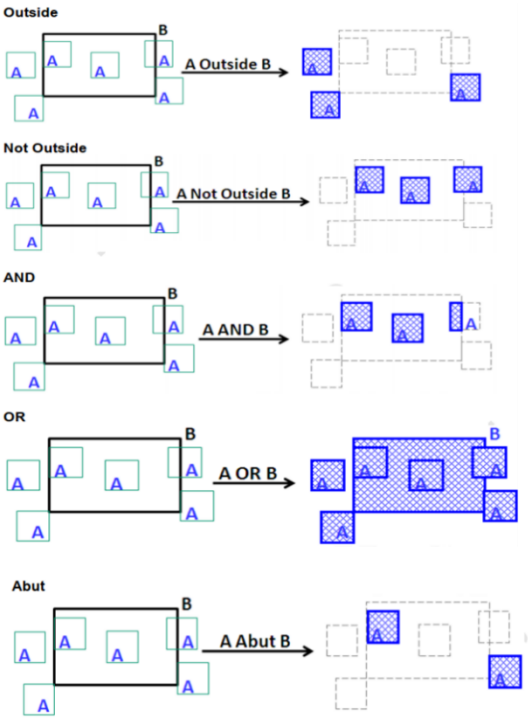
3.1 Layout Grid

This layout design grid --- 0.005 um

3.2 Definition of Layout Geometrical Terminology







3.3 Rule of interposer front

3.3.1 ALPAD

Rule Name	Description	Label	Op.	Value
rule.1	Width of ALPAD	A	>=	99
rule.2	Space of ALPAD	B	>=	189
rule.3	ALPAD enclosure of VIA0, tolerance is 0.01	C	>=	31
rule.4				

3.3.2 VIA0

Organic process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of VIA0, tolerance is 0.01	A	>=	54
rule.2	Space of VIA0	B	>=	54
rule.3	M1 enclosure of VIA0, tolerance is 0.01	C	>=	18
rule.4	VIA0 must inside M1			
rule.5	VIA0 must inside ALPAD			

3.3.3 DVIA0

Damascus process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of DVIA0, tolerance is 0.01	A	>=	27
rule.2	Space of DVIA0	B	>=	27
rule.3	DM1 enclosure of DVIA0, tolerance is 0.01	C	>=	13
rule.4	DVIA0 must inside DM1			
rule.5	DVIA0 must inside ALPAD			

3.3.4 Mn (1~3)

Organic process rules

Rule Name	Description	Label	Op.	Value
rule.1	Width of Mn	A	>=	10
rule.2	Space of Mn	B	>=	10
rule.3	Minimum {Mn or Mn_dummy} density across full chip		>=	18%
rule.4	Maximum {Mn or Mn_dummy} density across full chip		<=	63%

3.3.1 DMn (1~3)

Damascus process rules

Rule Name	Description	Label	Op.	Value
rule.1	Width of DMn	A	>=	9
rule.2	Space of DMn	B	>=	9
rule.3	Minimum {DMn or DMn_dummy} density across		>=	18%

	full chip			
rule.4	Maximum {DMn or DMn_dummy} density across full chip		<=	63%

3.3.2 VIAn (1~2)

Organic process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of VIAn, tolerance is 0.01	A	>=	18
rule.2	Space of VIAn	B	>=	27
rule.3	Mn enclosure of VIAn, tolerance is 0.01	C	>=	18
rule.4	VIAn must inside Mn			
rule.5	Mn+1 enclosure of VIAn, tolerance is 0.01	D	>=	13
rule.6	VIAn must inside Mn+1			

3.3.1 DVIAn (1~2)

Damascus process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of DVIAn, tolerance is 0.01	A	>=	10
rule.2	Space of DVIAn	B	>=	13
rule.3	DMn enclosure of DVIAn, tolerance is 0.01	C	>=	10.0
rule.4	DVIAn must inside DMn			
rule.5	DMn+1 enclosure of DVIAn, tolerance is 0.01	D	>=	11.0
rule.6	DVIAn must inside DMn+1			

3.3.2 RV

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of RV, tolerance is 0.01	A	>=	27
rule.2	Space of RV	B	>=	63
rule.3	Mtop enclosure of RV, tolerance is 0.01	C	>=	13
rule.4	RV must inside Mtop			
rule.5	UBM enclosure of RV, tolerance is 0.01	D	>=	18
rule.6	RV must inside UBM			

3.3.3 UBM

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of UBM, tolerance is 0.01	A	>=	45
rule.2	Space of UBM	B	>=	45
rule.3	Chip enclosure of UBM, tolerance is 0.01	C	>=	99

3.3.4 RVwafer

Only one design of RV or RVwafer design in customer product

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of RVwafer, tolerance is 0.01	A	>=	10
rule.2	Space of RVwafer	B	>=	16
rule.3	Mtop enclosure of RVwafer, tolerance is 0.01	C	>=	10.0
rule.4	RVwafer must inside Mtop			
rule.5	UBMwafer enclosure of RVwafer, tolerance is 0.01	D	>=	10.0
rule.6	RVwafer must inside UBMwafer			

3.3.5 UBMwafer

Only one design of UBM or UBMwafer design in customer product

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of UBMwafer, tolerance is 0.01	A	>=	13
rule.2	Space of UBMwafer	B	>=	13
rule.3	Chip enclosure of UBMwafer, tolerance is 0.01	C	>=	99

3.3.6 Mndummy (1~3)

Rule Name	Description	Label	Op.	Value
rule.1	Width of Mndummy	A	=	54
rule.2	Length of Mndummy	B	=	54
rule.3	Space of Mndummy	C	>=	27
rule.4	Space of Mndummy to Mn	D	>=	54
rule.5	Maximum {Mndummy OR Mn} density in window 1000 um x 1000 um, stepping 500 um		<=	18%
rule.6	Minimum {Mndummy OR Mn} density in window 1000 um x 1000 um, stepping 500 um		>=	9
rule.7	OFFSET of Mndummy in Y direction (for dummy fill only)	E	=	18
rule.8	OFFSET of Mndummy in X direction (for dummy fill only)	F	=	18

Mndummy.R.1[U] and Mndummy.R.2[U] can't checked by DRC, it for dummy fill deck only.

3.3.7 DMndummy (1~3)

Rule Name	Description	Label	Op.	Value

rule.1	Width of DMndummy	A	=	27
rule.2	Length of DMndummy	B	=	27
rule.3	Space of DMndummy	C	>=	13
rule.4	Space of DMndummy to DMn	D	>=	54
rule.5	Maximum {DMndummy OR DMn} density in window 1000 um x 1000 um, stepping 500 um		<=	18%
rule.6	Minimum {DMndummy OR DMn} density in window 1000 um x 1000 um, stepping 500 um		>=	9
rule.7	OFFSET of DMndummy in Y direction (for dummy fill only)	E	=	13
rule.8	OFFSET of DMndummy in X direction (for dummy fill only)	F	=	13

3.4 Rule of interpose backside

3.4.1 TSV

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of TSV, tolerance is 0.01	A	=	25/35
rule.2	Space of TSV if diameter of TSV is 20 um	B	>=	81
rule.3	Space of TSV if diameter of TSV is 30 um	B	>=	117
rule.4	Minimum TSV density across full chip		>=	9.0%
rule.5	Maximum TSV density across full chip		<=	9%

3.4.2 BMn (1~3)

Organic process rules

Rule Name	Description	Label	Op.	Value_Organic
rule.1	Width of BMn	A	>=	2
rule.2	Space of BMn	B	>=	2
rule.3	Minimum {BMn or BMn_dummy} density across full chip		>=	10%
rule.4	Maximum {BMn or BMn_dummy} density across full chip		<=	60%

3.4.1 DBMn (1~3)

Damascus process rules

Rule Name	Description	Label	Op.	Value

rule.1	Width of DBMn	A	>=	9
rule.2	Space of DBMn	B	>=	9
rule.3	Minimum {DBMn or DBMn_dummy} density across full chip		>=	18%
rule.4	Maximum {DBMn or DBMn_dummy} density across full chip		<=	63%

3.4.2 BVIAn (1~2)

Organic process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of BVIAn, tolerance is 0.01	A	>=	18
rule.2	Space of BVIAn	B	>=	27
rule.3	BMn enclosure of BVIAn, tolerance is 0.01	C	>=	18
rule.4	BVIAn must inside BMn			
rule.5	BMn+1 enclosure of BVIAn, tolerance is 0.01	D	>=	13
rule.6	BVIAn must inside BMn+1			

3.4.1 DBVIAn (1~2)

Damascus process rules

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of DBVIAn, tolerance is 0.01	A	>=	10
rule.2	Space of DBVIAn	B	>=	13
rule.3	DBMn enclosure of DBVIAn, tolerance is 0.01	C	>=	10.0
rule.4	DBVIAn must inside DBMn			
rule.5	DBMn+1 enclosure of DBVIAn, tolerance is 0.01	D	>=	11.0
rule.6	DBVIAn must inside DBMn+1			

3.4.2 BRV

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of BRV, tolerance is 0.01	A	>=	27
rule.2	Space of BRV	B	>=	63
rule.3	Mtop enclosure of BRV, tolerance is 0.01	C	>=	13
rule.4	BRV must inside Mtop			
rule.5	BUBM enclosure of BRV, tolerance is 0.01	D	>=	18
rule.6	BRV must inside BUBM			

3.4.3 BUBM

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of BUBM, tolerance is 0.01	A	>=	45
rule.2	Space of BUBM	B	>=	45
rule.3	Chip enclosure of BUBM, tolerance is 0.01	C	>=	99

3.4.4 BRVwafer

Only one design of BRV or BRVwafer design in customer product

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of BRVwafer, tolerance is 0.01	A	>=	10
rule.2	Space of BRVwafer	B	>=	16
rule.3	Mtop enclosure of BRVwafer, tolerance is 0.01	C	>=	10.0
rule.4	BRVwafer must inside Mtop			
rule.5	BUBMwafer enclosure of BRVwafer, tolerance is 0.01	D	>=	10.0
rule.6	BRVwafer must inside BUBMwafer			

3.4.5 BUBMwafer

Only one design of BUBM or BUBMwafer design in customer product

Rule Name	Description	Label	Op.	Value
rule.1	Diameter of BUBMwafer, tolerance is 0.01	A	>=	13
rule.2	Space of BUBMwafer	B	>=	13
rule.3	Chip enclosure of BUBMwafer, tolerance is 0.01	C	>=	99

3.4.6 BMndummy (1~3)

Rule Name	Description	Label	Op.	Value
rule.1	Width of BMndummy	A	=	54
rule.2	Length of BMndummy	B	=	54

rule.3	Space of BMndummy	C	>=	27
rule.4	Space of BMndummy to BMn	D	>=	54
rule.5	Maximum {BMndummy OR BMn} density in window 100 um x 100 um, stepping 50 um		<=	18%
rule.6	Minimum {BMndummy OR BMn} density in window 100 um x 100 um, stepping 50 um		>=	9
rule.7	OFFSET of BMndummy in Y direction (for dummy fill only)	E	=	18
rule.8	OFFSET of BMndummy in X direction (for dummy fill only)	F	=	18

BMndummy.R.1^{U} and BMndummy.R.2^{U} can't checked by DRC, it for dummy fill deck only.

3.4.7 DBMndummy (1~3)

Rule Name	Description	Label	Op.	Value
rule.1	Width of DBMndummy	A	=	27
rule.2	Length of DBMndummy	B	=	27
rule.3	Space of DBMndummy	C	>=	13
rule.4	Space of DBMndummy to DBMn	D	>=	54
rule.5	Maximum {DBMndummy OR DBMn} density in window 100 um x 100 um, stepping 50 um		<=	18%
rule.6	Minimum {DBMndummy OR DBMn} density in window 100 um x 100 um, stepping 50 um		>=	9
rule.7	OFFSET of DBMndummy in Y direction (for dummy fill only)	E	=	13
rule.8	OFFSET of DBMndummy in X direction (for dummy fill only)	F	=	13

BDMndummy.R.1^{U} and BDMndummy.R.2^{U} can't checked by DRC, it for dummy fill deck only.